

Projection Operator Definition

For $z \in \mathbb{R}^n$ and closed convex set K :

$$P_K(z) = \arg \min_{y \in K} \|z - y\|_2^2$$

Properties:

1. Non-expansive: $\|P_K(x) - P_K(y)\| \leq \|x - y\|$

2. Idempotent: $P_K(P_K(z)) = P_K(z)$

3. Characterization:

$$x^* = P_K(z) \text{ iff } (z - x^*)^T(y - x^*) \leq 0 \text{ for all } y \in K$$

Projection Lemma (Lemma 7.1)

For nonempty closed convex $K \subseteq \mathbb{R}^n$:

$$(x - P_K(z))^T(P_K(z) - z) \geq 0$$

for all $x \in K$ and $z \in \mathbb{R}^n$

Interpretation:

The vector from z to $P_K(z)$ makes

an obtuse angle with any direction from

$P_K(z)$ into K .